



### View the Space Station

Google's Street View now allows you to explore the International Space Station and all of its 15 modules. <http://dgit.in/ISSview>



### Self-balancing bike

If you've always wondered how Honda's self-balancing motor cycle works, here's your chance to understand: <http://dgit.in/hhsdmc>

external graphics card docks in the market and that's where we put the blame on Intel. Intel delivered a solid piece of technology by introducing the Thunderbolt standard and helped adoption in a much greater manner when it used the USB Type-C interface for Thunderbolt 3. USB, inarguably



PowerColor Devil Box

being the world's most popular consumer electronic interface would make Thunderbolt 3 ubiquitous instead of being the mainstay of only Apple devices. However, Intel had a steep charge for these Thunderbolt 3 chips. And even third party manufacturers had to pay a steep royalty to use the Intel IP. There's absolutely nothing wrong in charging for your own IP. The simple fact is that Intel's pricing caused manufacturers to stay away from Alpine Ridge whose controllers start from \$6.45 for the JHL6240 to \$8.55 for the JHL6540. Even ASMedia's own offerings cost around \$3. These prices may seem miniscule, but when you're building a motherboard with hundreds of components, these little prices stack up very quickly. So while Thunderbolt 3 became a very favourable interface in terms of bandwidth and flexibility, the high cost prevented it from being adopted widely. Some would even argue that if Intel had not adopted the USB Type-C standard, then Thunderbolt would have easily followed FireWire into obscurity. And earlier this year, Intel made its IP royalty-free. Essentially, manufacturers could license the technology once instead of paying royalties every time they used it, only if Intel agrees on these terms. This meant that more third-party manufacturers could

start making Thunderbolt 3 controllers at a much lower price point which, in turn, has led to a much greater adoption. Thunderbolt 3 being an active interface, requires the Thunderbolt 3 chip to be present at both ends of the cable. Hopefully, we'll start seeing more Thunderbolt 3 devices in the coming months.

### SO WHY THE SKEPTICISM?

This stems from the advancements that graphics cards have undergone over the last couple of years. They have become very power efficient, which makes them a lot easier to embed in their native form within laptops. Before Pascal came along, the GPUs that used to be embedded in laptops used to be downclocked versions of their desktop counterparts. Moreover, they weren't the exact same SKUs as the core counts were also very less. The usual power adapters for laptops can't provide enough juice and those that could were huge clunky monstrosities. These factors made it very difficult to include high powered GPUs in laptops but with Pascal and now with Polaris, we've started seeing the exact same desktop-grade GPUs in laptops.



GIGABYTE AORUS GTX 1070 Gaming Box

This begs the question – Why would you need graphics card docks anymore? After all, it's now feasible to have a GTX 1080 in a laptop thanks to the new Max-Q design specifications. Anyone who needs a powerful GPU might as well opt for convenience by getting a laptop with a powerful GPU already present. Especially since an external graphics card dock has

overheads like latency and bandwidth to worry about. Graphics intensive applications like games are affected by both these factors. Competitive games are highly optimised so most of them don't require powerful graphics but should there be a competitive video game with a need for a lot of visual compute power then the dock might not be the right solution.

### WHERE ARE WE HEADED THEN?

While this doesn't affect the other use cases for external docks, it sure does make the case weaker for graphics



HP Omen Accelerator

card docks. What's needed at this point is for the manufacturers of these docks to find a better way to sell them. We see a couple of use cases which should act in favour of these external graphics card docks. The foremost of them being the need for enterprise applications, carrying around a dock with a Quadro or Radeon Pro graphics card will seriously up your productivity and if you were to make use of the daisy chaining feature that the Thunderbolt 3 interface has, then you add more value to owning a dock rather than just being for gaming. Another use case is if the dock could be used not only for graphics cards but also for setting up an external RAID array by simply plugging in a daughter board with a RAID controller. These are some of the common and easy to implement applications that can help with greater adoption of the Thunderbolt 3 technology. And honestly, it's a little too early to ring the death knell for these docks. The coming months should, hopefully, shed more light on their future. 